

- 1 (a) Archimedes' Principle states that when a body is wholly or partially immersed in a fluid, it is buoyed by a vertical upward force (upthrust) equal in magnitude to the weight of fluid displaced by the body.

(b) (i) Upthrust due to one cube = weight of liquid displaced

$$= (0.2)(0.2)(1000)(9.81)x$$

Upthrust = weight of mass and plank

$$2(0.2)^2(1000)(9.81)x = (0.2)(9.81) + (10)(9.81)$$

$$x = 0.128m$$

- (ii) x decreases while y increases.

Upthrust on the raft remains unchanged

Taking moment about A, Upthrust on B has to increase as the mass shift to the right

(iii) *Max upthrust of B* $= (0.2)^3(1000)(9.81)$
 $= 78.5N$

Max upthrust of B is less than weight